

1. What structures pass through the jugular foramen?

- a. Cranial nerves VIII, IX, and X and the internal jugular vein**
- b. Cranial nerves IX, X, and XI and the internal jugular vein**
- c. Cranial nerves X and XI and the internal jugular vein**
- d. Cranial nerves IX and X and the internal jugular vein**
- e. Cranial nerves VIII and IX and the internal jugular vein**

2. You diagnose a patient with a strain pattern and decide to treat them with indirect balanced membranous tension. You examine the patient with a vault contact and treat them by exaggerating the freedoms by guiding the bony contacts of all fingers slightly cephalad while bringing your fingers together slightly on either side of the head. What cranial strain pattern are you treating?

- a. Extension**
- b. Flexion**
- c. Left torsion**
- d. Right torsion**
- e. Superior vertical shear**
- f. Inferior vertical shear**

Objectives Students will:

1. Review Landmarks. Assess for restrictions at various sutures and assess the PRM. **Motion test** for cranial strain patterns to decipher **strain patterns present** in our partners (flexion, extension, torsions, sidebending-rotations, vertical shears (superior or inferior), lateral strains, and/or SBS compression). Also learn to utilize **Unilateral Temporal Contact/Hold** to help Dx temporal dysfunctions.
2. Use **balanced membranous tension** (BMT) to bring the cranial mechanism back to a more physiologic state. Practice **Indirect BMT and/or Direct BMT** on your partners utilizing one or more cranial contacts (holds).
3. Assess the Occipito-Mastoid (OM) Suture for restriction. Importance? Consider temporal dysfunction, as well as cranial nerves IX, X, XI and jugular vein that exit near this region at the jugular foramen. Learn to treat this important area with **Directing the Tide and V-Spread of the OM Suture(s)**.
4. Learn about the wide-ranging utility of the **Compression of the 4th Ventricle (CV4)** technique, how to perform it, and how it relates to enhancing the motion and amplitude of the cranial rhythmic impulse, as well as its ability to augment the patient's healing capabilities. What might be the effect of CSF stasis on Circumventricular organs and CNS function? – consider therapeutic benefit of CV4 impact on basic physiologic centers including the brainstem, etc.
5. Review material from OMM lectures and labs 32 (Dr. Ventimiglia) and 33 (Dr. Terzella). Students will strive to diagnose and palpate various cranial strain patterns as well as demonstrate OA & condylar decompression and the various lift techniques from antecedent labs.

Students will continue to appreciate the complex structure and function of skeletal, arthrodial, myofascial, vascular, lymphatic, and CNS elements as a whole in the context of dxing & treating with OCF/ OCMM- osteopathic cranial manipulative medicine.

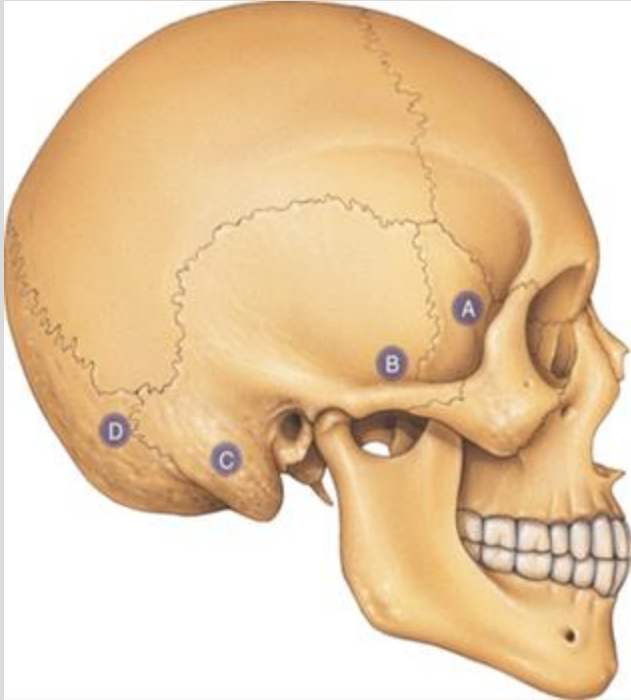
Assess, “Dur-assess” and Reassess

- Note similar to other body parts, that for all the cranial and facial techniques, one must assess for somatic dysfunction (i.e. TART- such as tenderness, soft tissue edema, muscle hypertonicity of the face or neck, restricted Frontal or Zygomatic bone motion, etc.)
- During OMT or OCF one can dur-assess and become more proficient at establishing whether this treatment is the best avenue to accomplish the intended result. (or should we modify? Try a different approach? Would direct or indirect treatment work best in this patient today? Should I go lighter or add a little more force? Etc.)
- After OMT, one should reassess by palpating the treated regions or **re-performing the technique** to see if the treated structures feel freer? Did the treatment seem to make a change in the degree of SD? In practice, we can also ask how a patient feels (subjective improvement may not be immediate, however). One can reassess by rechecking for changes in TART.

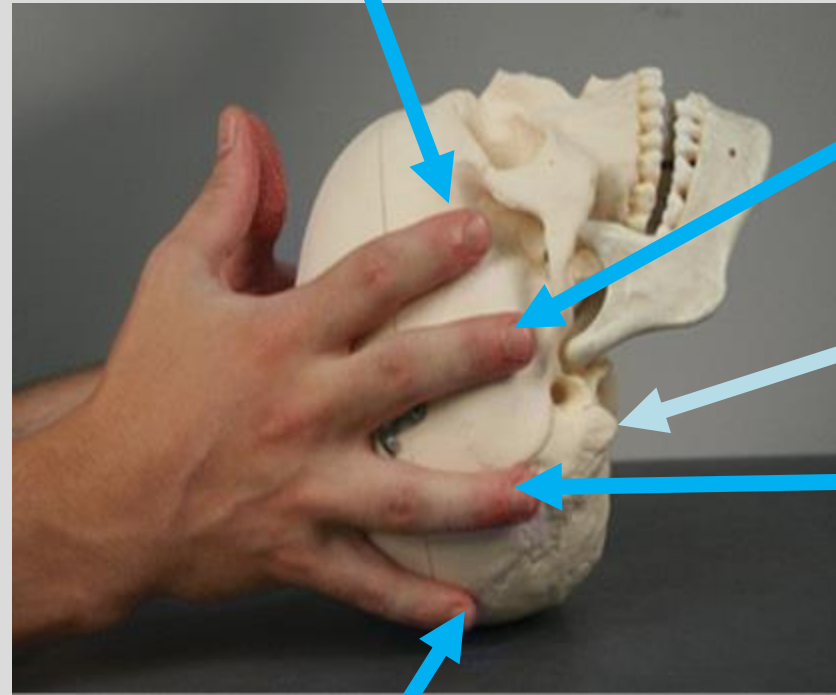
Vault Contact (a.k.a. Vault Hold)

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greater wings of the sphenoid



temporal bone
anterior to the ear

mastoid process

temporal bone
posterior to the ear

squamous portion of the occiput

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Review of Physiologic Strain Patterns

Pattern	Number and Direction of Axis	Movement of Occiput & Sphenoid relative to each other	Physiologic or non-physiologic Pattern	Description of air hands from vault contact
Flexion	2 transverse axes	Move around transverse axes in opposite directions	Physiologic	"Swelling" The head expands into the operator's hands laterally and shortens vertically, so palms move laterally, fingers spread & Fingers move towards the feet
Extension	2 transverse axes	Move around transverse axes in opposite directions	Physiologic	"Receding" Opposite as above: Palms come closer to one another and cephalad as fingers approximate
Torsions (R or L)	1 A/P axis	Move around A/P axis in opposite directions	Physiologic	Hands rotate (twist) in opposite directions (one twists sup-posterior, one inf-anterior) Named for side of high GW.
Sidebending & Rotation (R or L)	2 parallel vertical axes, 1 A/P axis	Sidebending: Move in opposite directions around parallel vertical axes Rotation: Move in same direction around A/P axis	Physiologic	The hand on the side of convexity moves caudally with fingers spreading apart. The hand on side of concavity moves cephalad with fingers coming together. Named for side of fullness (convexity).

Review of Non-Physiologic Strain Patterns

Pattern	Number and Direction of Axis	Movement of Occiput & Sphenoid relative to each other	Physiologic or non-physiologic Pattern	Description of air hands
Lateral Strain (R or L)	2 parallel vertical axes	Move around vertical axes in same direction	Non-physiologic	Hands slant. Index fingers on GW of sphenoid move opp to direction of pinkies on occiput. BUT <u>Named for direction of basisphenoid</u> (which is the same direction as the pinkies, opp the direction of the GW)
Superior Vertical Strain/Shear	2 transverse axes	Move around transverse axes in same direction	Non-physiologic	Fingers point down towards feet (wrists ulnar-deviate). <u>Named for direction of basisphenoid</u>
Inferior Vertical Shear/Strain	2 transverse axes	Move around transverse axes in same direction	Non-physiologic	Fingers point up toward head. (wrists radial-deviate) <u>Named for direction of basisphenoid</u>
Sphenobasilar Synchronosis Compression	No axes	N/A	Non-physiologic	Feels rock-hard and heavy like a bowling ball head

Motion Testing for Strain Patterns

- While in the vault contact, palpate the cranial rhythmic impulse (CRI) as the head moves into flexion and extension.
- During a flexion/inhalation phase, very gently (and by gently, we mean to just think about performing the motion rather than actually doing it) guide the head into the strain pattern you would like to test for. i.e. Right torsion. See how the head responds. Then relax your intention and allow the head to go through a full phase of flexion and extension. As a 2nd flexion phase begins, test for the opposite direction. i.e. Left torsion.
- Compare sides.

Motion Testing for Strain Patterns

(continued)

- Repeat for all the strain patterns.
- The direction that has more motion (the freedom) is the one the dysfunction(s) is/ are named for.
- Remember to be gentle with motion testing. You do not want to induce a strain pattern.

Treating cranial strains with two kinds of Balanced Membranous Tension (BMT)

1. **Direct BMT** – Gently guide the strain pattern present into its barriers, i.e. opposite to the way that it “wants to go.” Hold point of balanced tension (POBT) until you feel a release(s).
 - Commonly used in infants and children before sutures fully develop
 - Can be used when indicated for trauma
 - Can be used for articular restrictions
 2. **Indirect BMT** – Gently guide the strain pattern into its freedoms, i.e. the way it “wants to go.” Hold point of balanced tension (POBT) until you feel a release(s).
- Exaggeration and Disengagement are two other technique classifications that you could read about but will not be discussed in lab, nor will you be held responsible for them at this time.

Summary Slide

1. You have located various cranial **landmarks** and reviewed and practiced sensing the **CRI and PRM utilizing cranial contacts** (holds) including a new contact: **Unilateral Temporal Hold**.
2. You will have reviewed **cranial strain patterns** and motion tested for them on your partner
3. You will have treated the strains present using **indirect and/or direct BMT** technique
4. You will have practiced **V-spread and directing the tide** to release the O-M sutures AND have attempted to apply the **CV-4** technique to your lab partner.
5. Please use your hands and mind to continue to integrate your understanding of anatomy, physiology and pathology in your understanding of health and various dis-ease states.